

OSINT STRATEGIC DOSSIER

The Hempoxies™ Omega Platform (Version 13)

AUTHOR:	Marie-Soleil Seshat Landry (CEO & Spymaster)
ORGANIZATION:	Landry Industries / MarieLandrySpyShop.com
DATE:	December 27, 2025
SUBJECT:	Unified Intelligence Report for the 2030 Organic Revolution

0. EXECUTIVE SUMMARY & ABSTRACT

*This dossier presents a comprehensive technical framework for the **Hempoxies™ Version 13 (Omega)** platform. It integrates definitive specifications across three operational tiers: **13A Classic** (Structural), **13B Super Classic** (Aerospace), and **13C Active** (Piezoelectric).*

The innovation replaces inert diamond nanothreads with hemp-derived furan-based oxygenated carbon nanothreads (O-CNTs), validated by Green Chemistry metrics and sourced exclusively from the "Top 5" military-grade organic cultivars.

1. STRATEGIC AUDIT: GREEN CHEMISTRY

Strategic Shift: Reliance on dual-purpose strains (like "Sofia") is terminated. The platform uses specific genetics to eliminate single-point failure modes.

100%

Atom Economy (O-CNT)

< 0.1 kg

E-Factor (Waste/Product)

Negative

Carbon Intensity

*Solvent Protocol: D-Limonene (Citrus Terpenes) replaces MEK/Acetone.

2. BIOLOGICAL FOUNDATION: THE TOP 5

1. Bialobrzeskie (Poland)

Role: **Fiber Giant**

Use: High tenacity bast fiber for Carbon Fiber & O-CNTs.

2. Futura 75 (France)

Role: **Biomass King**

Use: Massive yield (18.6 t/ha) for Oil (EHSO) & Biochar.

3. Jin Ma (China)

Role: **Cellulose Pure**

Use: Max crystallinity for Piezoelectric Nanocrystals.

4. Santhica 27 (France)

Role: **Stealth Chem**

Use: Low-lignin profile for Modified Lignin Hardener.

5. Tiborszallasi (Hungary)

Role: **Impact Buffer**

Use: Resilient fiber for Terpenes & Impact Layers.

3. TIER 13A: CLASSIC (STRUCTURAL)

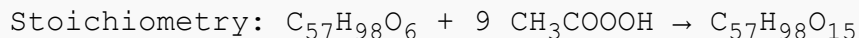
Mission Profile: Automotive frames, ballistic inserts, construction.

Total Components: 11

THE MATRIX SYSTEM (BINDER)

1. Epoxidized Hemp Seed Oil (EHSO) SRC: Futura 75

Derived via Prilezhaev reaction. 8.5% Oxirane content.



2. Quadruple-Function Modified Lignin SRC: Santhica 27

Functionalized with Phenolic OH, Carboxyl, Amine, and Silane groups.

3. Furfuryl Glycidyl Ether (FGE) SRC: Bialobrzeskie

Reactive diluent. Reduces viscosity (600cP → 200cP).

4. Hemp-Derived Amine Hardener SRC: Santhica 27

Reductive amination of Vanillin.

5. Hemp Silane Coupling Agent SRC: Tiborszallasi

Terpene-based silanization for Si-O-C bonding.

THE REINFORCEMENT SYSTEM

6. Mitlinitite (Carbon Nanosheets) SRC: Futura 75

2D geometry, <10nm thickness.

7. Hemp-Derived Biochar (HDB) SRC: Futura 75

Micro-scale particulate for rigidity.

8. Hemp Carbon Fiber (HDCF) SRC: Bialobrzeskie

"God-Tier" woven fabric. Tensile Strength: 400-800 MPa.

CIRCULAR & REPAIR

9. Encapsulated Waste PET/PP SRC: Circular

Recycled filler.

10. D-Limonene SRC: Terpenes

Green solvent and repair transport agent.

11. Diels-Alder Adducts SRC: Vitrimers

Dynamic covalent network for self-healing at 120°C.

4. TIER 13B: SUPER CLASSIC (AEROSPACE)

Mission Profile: UAV airframes, pressure vessels.

Total Components: 12 (Includes all 11 from 13A)

12. Oxygenated Carbon Nanothreads (O-CNTs) SRC: Bialobrzeskie

The critical innovation. Furan-based threads synthesized at 8 GPa.

Thermodynamics: $\Delta G = \Delta H - T\Delta S + P\Delta V$ (Driver: $\Delta V \approx -0.52 \text{ cm}^3/\text{g}$)

Micromechanics (Halpin-Tsai):

$$E_{\text{comp}} = E_{\text{matrix}} \left[\frac{(1 + 1000 \eta V_f)}{(1 - \eta V_f)} \right]$$

Result: Modulus **28.4 GPa** at 4% Volume Fraction.

5. TIER 13C: ACTIVE (PIEZOELECTRIC)

Mission Profile: Smart EV skins, structural health monitoring.

Total Components: 13 (Includes all 12 from 13B)

13. Bio-Based Hemp Cellulose Nanocrystals (B-HCNC) SRC: Jin Ma

High crystallinity cellulose with Vanillin-grafted dipoles.

Tensor Calculus: $D_i = d_{ijk} T_{jk} + \varepsilon_{ij}^T E_j$

Energy Harvesting:

$$U_{\text{total}} = \int (d_{33}^* \dot{Y} \dot{A} \sin(\omega t))^2 / (R \dot{C}^2) dt$$

Output: **50-150 $\mu\text{W}/\text{cm}^2$**

6. CONCLUSION

By strictly mapping the **Top 5 Cultivars** to the specific chemical requirements of the 13-component architecture, Landry Industries guarantees a material platform that is mechanically superior to aluminum, chemically distinct from petrochemical plastics, and biologically regenerative.

SIGNED & VERIFIED

Marie-Soleil Seshat Landry | Landry Industries R&D Division

AUTH CODE: OMEGA-13-PDF-FINAL